

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2212

Flambeau Solar Partners 100 kV
December 22, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2212. The Interconnection Customer proposes a 57.6 MW battery energy storage facility interconnecting at the Flambeau Solar Partners 100 kV substation in the Illinois area. The request was received on October 23, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 681 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$114,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Pineville Delivery 1 - Kimberly 100.0 kV Line
Contingency: Pineville Delivery 1 - Kimberly 100.0 kV
Loading: 101.8% (Rating: 179 MVA)
DFAX: 7.8%
Pre-project loading: 99.3%
- Facility: Kimberly - Pineville Delivery 1 100.0 kV Line 2
Contingency: Kimberly - Pineville Delivery 1 100.0 kV Circuit 1
Loading: 101.8% (Rating: 179 MVA)
DFAX: 7.8%
Pre-project loading: 99.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pineville Delivery 1 - Kimberly 100.0 kV Line	Reconductor line	\$634,000	9% (\$56,000)
Kimberly - Pineville Delivery 1 100.0 kV Line 2	Reconductor line	\$654,000	9% (\$58,000)
	Total allocated to project		\$114,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.